

Small language model

Small language models or **compact language models** are artificial intelligence language models designed for human natural language processing including language and text generation. They are smaller in scale and scope than large language models.

A *large* language model typically contains hundreds of billions of training parameters, with some models exceeding a trillion parameters. This substantial parameter count enables the model to encode vast amounts of information, thereby improving the generalizability and accuracy of its outputs. However, training such models demands enormous computational resources, rendering it infeasible for an individual to do so using a single computer and graphics processing unit.

Small language models, on the other hand, use far fewer parameters, typically ranging from a few thousand to a few hundred million. This make them more feasible to train and host in resource-constrained environments such as a single computer or even a mobile device.^{[1][2][3][4][5]}

Most contemporary (2020s) small language models use the same architecture as a large language model, but with a smaller parameter count and sometimes lower arithmetic precision. Parameter count is reduced by a combination of knowledge distillation and pruning. Precision can be reduced by quantization. Work on large language models mostly translate to small language models: pruning and quantization are also widely used to speed up large language models.

Models

Some notable models are:^[2]

- Below 1B parameters: Llama-Prompt-Guard-2-22M (detects prompt injection and jailbreaking, based on DeBERTa-xsmall), SmolLM2-135M, SmolLM2-360M
- 1–4B parameters: Llama3.2-1B, Qwen2.5-1.5B, DeepSeek-R1-1.5B, SmolLM2-1.7B, SmolVLM-2.25B, Phi-3.5-Mini-3.8B, Phi-4-Mini-3.8B, Gemma3-4B; closed-weights ones include Gemini Nano
- 4–14B parameters: Mistral 7B, Gemma 9B, Phi-4 14B.

Phi-4 14B is marginally "small" at best, but Microsoft does market it as a small model.^[6]

Language model with small pre-training dataset

Traditional AI language systems need enormous computers and vast amounts of data.

Pre-training matters, even tiny models show significant performance improvements when pre-trained performance increases with larger pre-training datasets. Classification accuracy improves when pre-training and test datasets share similar tokens.

Shallow architectures can replicate deep model performance through collaborative learning.^[7]

See also

- [Edge computing](#)

References

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